Design of Unidirectional Security Gateway System for Secure Monitoring of OPC-UA Data

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***Abstract*—A unidirectional security gateway is a network system in which data can travel in only one direction. Therefore, the system are used in protecting the safety and reliability of various critical infrastructures. Most of all, it has to be deployed with send and receive transceivers disconnected for one direction to ensure information security. In this paper, we introduce a unidirectional security gateway system, named UNIWAY that has a unidirectional data transfer mechanism for secure monitoring of OPC-UA data. The proposed system has the advantage of monitoring the data efficiently while protecting the OPC-UA server in the critical infrastructure.**

***Keywords*— Unidirectional security gateway, UNIWAY, OPC-UA, cyber-attack.**

I. Introduction

A control system is a computer based system that is widely used in typical factories and in national infrastructures. The continuously increasing convergence of the control system and IT (Information Technology) is the growing possibility of cyber-attacks on control systems [1]. That is, cyber security has been a topic of concern for a wide variety of critical infrastructures for over a decade [2]. It was initially thought that critical infrastructures were safe from cyber-attack because they use an isolated network and a closed control protocol, but cyber threats such as Stuxnet, Duqu and Flame have become the key issue of late. Therefore, the development of security technology customized to a critical infrastructure is required.

A unidirectional data transfer technology is a technique that prevents important data from being leaked due to external hacking and security-related accidents. Recently, it has been introduced into a separate closed network such as a control network to enhance security. It denotes one of inter-network data transmission technologies, and blocks a physical line that enables data to be transmitted from an external network to a separate closed network, thus completely deleting the external threat path itself. A unidirectional security gateway with such a technique is a network system in which data can travel in only one direction. The systems are designed to securely integrate a variety of applications by replicating servers from a protected internal network to external networks [3]. Also, the systems are deployed with send and receive transceivers removed or disconnected for one direction [4]. Therefore, the systems [5-7] have an advantage that can transmit information out of the critical infrastructures into an external network without any risk of cyber-attacks.

Most of the control networks used in a critical infrastructure have the manner of monitoring the data of the separated closed network in the external network, which is performed through control protocols of various application layers suitable for the control network. That is, to apply the unidirectional security gateway system to the control network, it is possible through the respective proxy technology for various control protocols such as Modbus, DNP3 and PROFINET. Among these control protocols, the OPC-UA protocol is the latest OPC standard for interoperability between different control protocols [8]. Therefore, this paper firstly considers the unidirectional data transfer mechanism for OPC-UA data.

This paper introduces a unidirectional security gateway, named UNIWAY (UNIdirectional gateWAY), and describes a secure monitoring scheme of OPC-UA data in the proposed system. The rest of this paper is organized as follows: Chapter 2 presents the architecture and detailed mechanism of the proposed system; Chapter 3 discusses the conclusion and future plan.

II. Design of the Proposed System

A. Deployment of the Proposed System

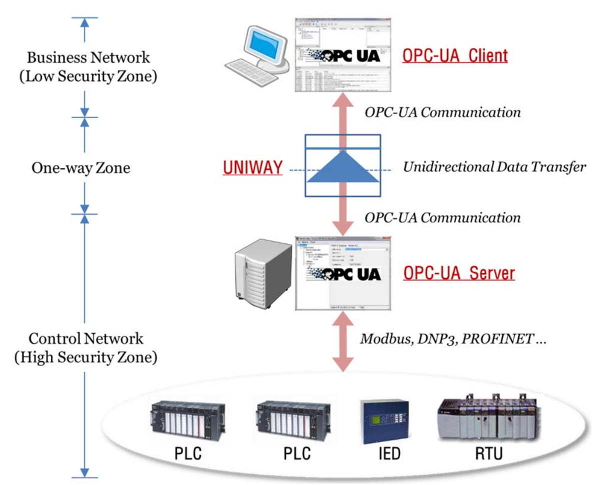

Figure 1. Deployment of UNIWAY system

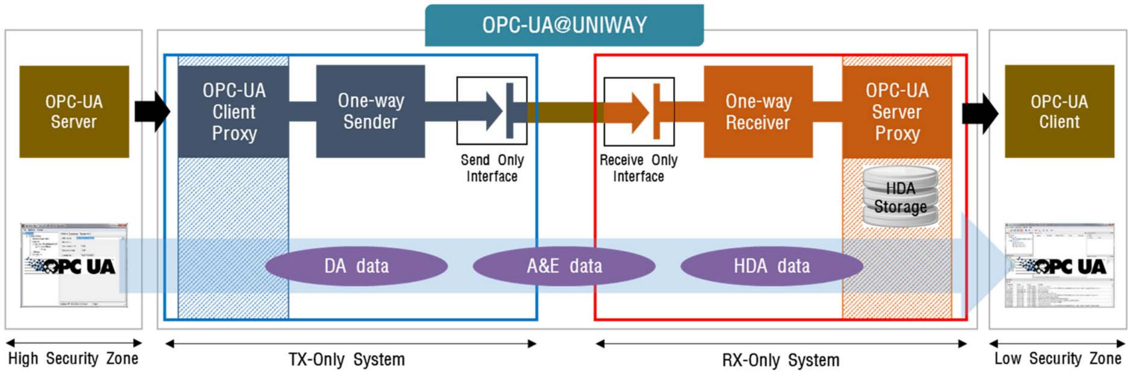

Figure 2. Architecture of UNIWAY System

UNIWAY system provides a unidirectional data transfer to critical infrastructures; it is positioned between the business network and the control network. Fig. 1 shows the typical deployment for secure transfer of OPC-UA data. Our system allows information to leave a protected internal network such as control network, and blocks all connections originating on external networks such as business network. Here, the internal network is referred to as a high security zone, and the external network is referred to as a low security zone. As shown in the figure, our system is located between the OPC-UA server and OPC-UA client and performs OPC-UA communication on both sides. In other words, it provides a proxy function for OPC-UA server and client.

B. *Overall Architecture*

Our system supports a secure monitoring of OPC-UA data in the above unidirectional data transfer environment. Fig. 2 shows the overall architecture for a unidirectional data transfer between OPC-UA server and OPC-UA client. As shown in the figure, it consists of a TX-Only system and a RX-Only system. Each system consists of functional blocks that perform symmetrical functions, each of which is described as follows:

- OPC-UA Client Proxy (or Send Proxy): it emulates the original OPC-UA client, and communicates the OPC-UA server in the high security zone.
- OPC-UA Server Proxy (or Receive Proxy): it emulates the original OPC-UA server, and communicates the OPC-UA client in the low security zone. Here, it has a repository for storing large amounts of HDA (Historical Data Access) data.
- One-way Sender/Receiver: it encodes/decodes the unidirectional data being transferred between a TX-Only system and a RX-Only system.
- Send/Receive Only Interface: each interface has a fiber-optic communications, and only contains a fiber-optic transmitter or receiver according to the data transfer direction.

Using the above functional blocks, the proposed system can effectively support a secure monitoring of all OPC-UA data such as DA (Data Access) data, A&E (Alarm & Event) data and HDA data.

C. *Unidirectional Data Transfer Mechanism*

Fig. 3 shows an overall procedure for transmitting OPC-UA server data (in a high security zone) to an OPC-UA client (in a low security zone) through a TX-Only/RX-Only system equipped with an OPC-UA based send/receive proxy. As shown in the figure, unidirectional data transfer for secure monitoring of OPC-UA data is largely performed in the order of three operations. Details of each operation are as follows.

1) OPC-UA Server Proxy Establish: it is required to construct a server proxy by receiving the node data managed by the OPC-UA server in a high security zone. So, the send proxy connects the OPC-UA server in the high security zone, requests and receives node information, and transmits the node information to the receive proxy to construct an OPC-UA server proxy. In this case, the OPC-UA server proxy uses the same IP address as the OPC-UA server in the high security zone, so that the OPC-UA client in the low security zone has the same effect as directly connecting to the OPC-UA server in the high security zone.

2) Real-Time Data Update: it is performed to update the real-time data of the OPC-UA server proxy established through the above processing. The real-time data of OPC-UA is composed of DA data and Alarm & Event data based on pre-configured node data. That is, the OPC-UA send proxy requests and receives all the real-time data provided by the OPC-UA server in the high security zone, and immediately delivers it to the receive proxy. The receive proxy continuously updates the real-time data of the managed node built on the received data, and provides data suitable for the request of the OPC-UA client in the low security zone. Here, the real-time data request is performed only once, and the requested real-time data is automatically received whenever the data of the OPC-UA server changes.

3) Historical Data Update: it is done to update the HDA data of the OPC-UA server proxy and is performed in the same manner as the above real-time data update. However, unlike real-time data that is automatically updated, historical data is transmitted only on request. Therefore, it is performed according to the periodic iteration according to the setting of the send proxy.

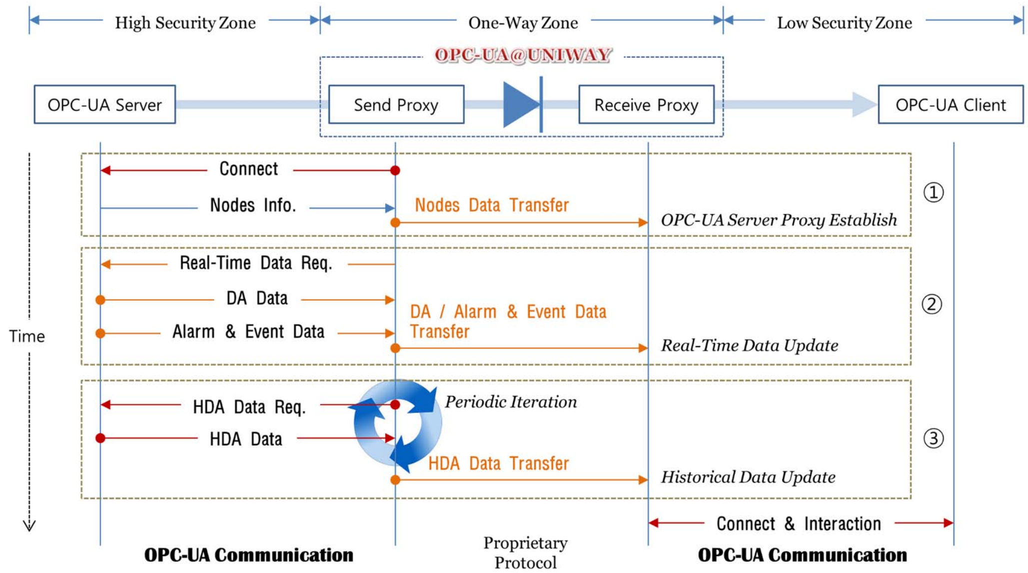

Figure 3. Overall Procedure for Unidirectional Transfer of OPC-UA data

Through the above operations, the UNIWAY system provides a secure OPC-UA data monitoring technique. In other words, by providing an OPC-UA data monitoring method suitable for a unidirectional data transfer environment, it provides a technique of securely transmitting OPC-UA data to a low security zone while blocking external intrusion to a high security zone which is a separate closed network. That is, the above mechanism provides a technique for effectively monitoring in the low security zone while protecting the OPC-UA server in the critical infrastructure.

III. Conclustions

This paper has introduced the design of a unidirectional security gateway called UNIWAY, and describes a secure monitoring scheme of OPC-UA data in the proposed system. The proposed system supports a secure monitoring of all OPC-UA data such as DA data, Alarm & Event data and HDA data in a unidirectional data transfer environment. The scheme is provided through the proxy function for the original OPC-UA server and client, and basically performed with send and receive transceivers removed or disconnected for one direction, in guaranteeing information security. Therefore, the proposed system has the advantage of monitoring the data efficiently while protecting the OPC-UA server in the critical infrastructure.

In the future, we plan to address the problems identified by various tests. In addition, we will expand the functionality to support other protocols in addition to the OPC-UA protocol.

Acknowledgment

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